

Ministerul Educației și Cercetării al Republicii Moldova

Universitatea Tehnică a Moldovei

Facultatea Calculatoare, Informatică și Microelectronică

Technical Report:
Mercantilism Simulation
Economic Model

Elaborated:

st. gr. FAF-243 Kushnirenko Ecaterina

Verified:

prof. univ. „

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1 OVERVIEW

1.1 Objective

The objective of this simulation is to model four competing IT-sector nations operating in a **mercantilist** trade environment and to compare outcomes against a **free-trade** alternative. Each nation produces one raw resource and must import the other three through trade, while building navies, forming alliances, imposing tariffs, and engaging in espionage. The central academic thesis under investigation is:

Free trade produces higher global wealth (positive-sum) while mercantilism creates winners and losers through zero-sum redistribution.

The simulation runs **24 ticks** (trading seasons) per scenario, each tick lasting **12 real seconds**. Both scenarios run back-to-back and final global wealth totals are compared.

1.2 Nations

Four nations participate in the simulation, each associated with a primary resource, a map position, and a team color.

Table 1: Nation Configuration

Nation	Primary Resource	Color	Map Position
Endava	Ore	Dark blue (0, 36, 125)	Top-left (−160, 1, −160)
Amdaris	Herbs	Blue (0, 85, 164)	Top-right (160, 1, −160)
GridDynamics	Meat	Red (170, 21, 27)	Bottom-left (−160, 1, 160)
Globant	Logs	Amber (255, 174, 0)	Bottom-right (160, 1, 160)

1.3 Starting State

Each nation begins tick 0 with the following attributes:

Table 2: Initial State per Nation (Tick 0)

Attribute	Value
Wealth	1,000g
Trade ships	2
Warships	1
Own resource stock	70 units
Other resource stocks	40 units each
Economy tier	RAW (1)
All bilateral relations	50 (neutral)
Diplomatic state	NEUTRAL (all pairs)
Active tariffs	none
Active privateers	0

2 TICK EXECUTION ORDER

Every tick the following steps execute **in this exact sequence**:

1. Update economy tiers (wealth threshold check)
2. Produce & consume resources (+20 own, -4 all)
3. Resource-need relation boosts (warm toward needed suppliers)
4. Animate ships (visual, fire-and-forget)
5. Random market events (40% chance)
6. Tariff AI decisions (mercantilist only, tick ≥ 2)
7. Diplomatic AI decisions (mercantilist only, every 2 ticks)
8. Calculate trade (exports, imports, resource transfer)
9. Resolve plunder (warship raids + animations)
10. Resolve privateers (corsair interceptions)
11. Resolve sabotage (harbour sabotage attempts)
12. Update navy sizes (Richardson arms race)
13. Relationship maintenance (alliance boost, embargo drain, decay)
14. Apply wealth changes ($W = W + \text{exports} - \text{imports} + \text{plunder} - \text{navy}$)
15. Apply degradation (fleet losses, export penalties)
16. Allied foreign aid (critical/bankrupt ally rescue)
17. Record tick snapshot (CSV data collector)
18. Broadcast to clients (WealthUpdated RemoteEvent)

3 WEALTH FORMULA

3.1 Core Equation

The wealth of nation i at the end of tick t is given by:

$$W_i(t) = W_i(t - 1) + X_i - M_i + P_i - C_i \quad (1)$$

Table 3: Wealth Formula Symbols

Symbol	Meaning
$W_i(t)$	Nation i 's wealth at end of tick t
X_i	Export income earned this tick
M_i	Import spending this tick
P_i	Plunder gained (raids + privateers)
C_i	Navy maintenance cost

Wealth is floored at **0** — nations cannot go negative. Plunder *losses* are applied directly to `nation.wealth` inside `NavalSystem.resolvePlunder()` before step 14. The `wealth_delta` CSV column reflects only gains ($X - M + P_{\text{gained}} - C$); subtract `plunder_lost` to get the true net change.

3.2 Trade Net (Typical Healthy Nation, RAW Tier)

Mercantilist scenario:

exports	= 53.3g x 3 routes x 0.74 distance	~ = 118g
imports	= 118 x 0.55	~ = 65g
navy	= 1 warship x 30g	~ = 30g
net	= 118 - 65 - 30	= +23g/tick

Free-trade scenario:

exports	= 118 x 1.65	~ = 195g
imports	= 195 x 0.55	~ = 107g
navy	= 20g (flat cost, no arms race)	
net	= 195 - 107 - 20	= +68g/tick

4 RESOURCE & TRADE SYSTEMS

4.1 Resource Production and Consumption

Each tick, before trade executes, stocks are updated as follows:

$$\text{own_stock} += 20 \quad (\text{capped at } 200) \quad \text{all_stocks} -= 4 \quad (\text{floored at } 0) \quad (2)$$

Table 4: Buying Behavior by Stock Level

Stock Level	Buying State	Price Modifier
< 15 (urgent)	Actively seeks resource	$\times 1.5$
15 – 199 (normal)	Willing to buy at base price	$\times 1.0$
≥ 200 (saturated)	Will NOT buy — route income blocked	—

Units delivered per trade route per tick:

$$\text{units_delivered} = \min\left(6 \times \frac{2}{3}, \text{seller_stock}\right) = \min(4, \text{seller_stock}) \quad (3)$$

4.2 Export Income Formula

For each non-embargoed partner, route income is calculated as:

$$\begin{aligned} \text{route_income} = & \text{base_price} \\ & \times \text{urgency_premium} \\ & \times [+ \text{manufactured_price} \times \text{shipsPerRoute}] \quad (\text{tier} \geq 2) \\ & \times [+ \text{technology_price} \times \text{shipsPerRoute}] \quad (\text{tier} \geq 3 \wedge \text{allied}) \\ & \times \text{relationship_modifier} \times \text{free_trade_multiplier} \\ & \times \text{alliance_bonus} \times (1 - \text{tariff_penalty}) \times \text{distance_efficiency} \end{aligned}$$

where $\text{base_price} = 160/3 \approx 53.3\text{g}$ per route.

Table 5: Trade Multipliers Summary

Factor	Value	Condition
Urgency premium	$\times 1.5$	Partner stock < 15
Manufactured goods	$\text{base} \times 2.2 \times \text{ships}$	Seller tier ≥ 2 , buyer tier < 2
Technology	$\text{base} \times 4.0 \times \text{ships}$	Seller tier ≥ 3 AND allied
Relationship bonus	$\times (1 + (\text{rel} - 50) \times 0.003)$	Relation > 50; max +15% at 100
Relationship penalty	$\times \max(0.3, 1 - (\text{rel} - 50) \times 0.005)$	Relation < 50; max –35% at 0
Free trade bonus	$\times 1.65$	Free trade scenario only
Alliance bonus	$\times 1.30$	Allied diplomatic state
Tariff penalty	$\times 0.50$	Partner has active tariff on you
Distance efficiency	$1/(1 + d/320 \times 0.3)$	Always applied
Degradation penalty	$\times 0.80/0.55/0.30$	Struggling / Critical / Bankrupt

4.3 Import Spending

$$M_i = X_i \times 0.55 \quad (4)$$

If the nation has active tariffs (mercantilist only), import costs decrease:

$$\text{reduction} = \min(0.80, \text{TARIFF_RATE} \times \text{tariff_count} \times 0.4) \quad M_i = M_i \times (1 - \text{reduction}) \quad (5)$$

5 ECONOMY TIER SYSTEM

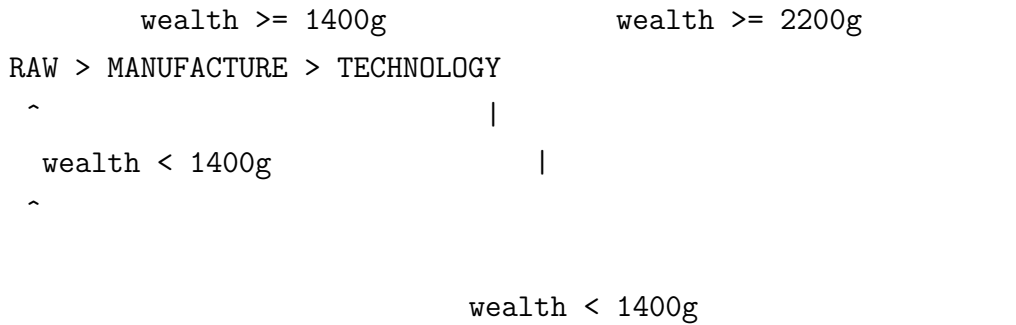
5.1 Tier Definitions

Table 6: Economy Tier Thresholds and Effects

Tier	Name	Wealth Required	Exports	Multiplier
1	RAW	0g	Raw resources only	×1.0
2	MANUFACTURE	≥ 1,400g	Raw + Manufactured goods	×2.2 on MFG
3	TECHNOLOGY	≥ 2,200g	Raw + MFG + Technology	×4.0 on TECH

Tier is **recalculated every tick**. A wealth drop below a threshold causes **immediate regression** — there is no hysteresis. Technology exports are restricted to allied routes only, reflecting the diplomatic trust required for advanced knowledge transfer.

5.2 Tier State Machine



6 RELATIONSHIP & DIPLOMACY SYSTEMS

6.1 Relationship Structure

Every pair (A, B) has two independent scores: $A \rightarrow B$ and $B \rightarrow A$, both on a 0–100 scale, initialized at **50** (neutral). The trade modifier uses the *average* of both directions; diplomatic AI uses each nation’s own view of the other.

6.2 Relationship Modifiers per Tick

Symmetric changes (both directions change equally):

Table 7: Symmetric Relationship Events

Event	Delta	When
Successful trade with partner	+3	After trade calculation
Delivered urgently needed resource	+3	Partner stock < 30 of that resource
Being allied	+5	End-of-tick maintenance
Urgently needs partner’s resource	+4	Stock < 15, after production step
Moderately low stock of partner’s good	+1	Stock < 30
Embargo active	−1	End-of-tick maintenance
Embargoed by needed supplier	−6	Stock < 15 + embargoed
Both saturated + not allied	−1	When both stocks ≥ 180
Allied foreign aid given	+8	After aid transfer
Natural decay toward 50	$\pm 1/\text{tick}$	Each direction independently

Asymmetric changes (aggressor and victim experience differently):

Table 8: Asymmetric Relationship Events

Event	Aggressor	Victim	When
Raid / plunder	−3	−10	After successful plunder
Sabotage (success)	−8	−30	After sabotage success
Sabotage (fail)	−8	−15	After uncovered failure
Tariff imposed	−3	−12	When tariff is set

6.3 Relationship Thresholds

Table 9: Action Thresholds

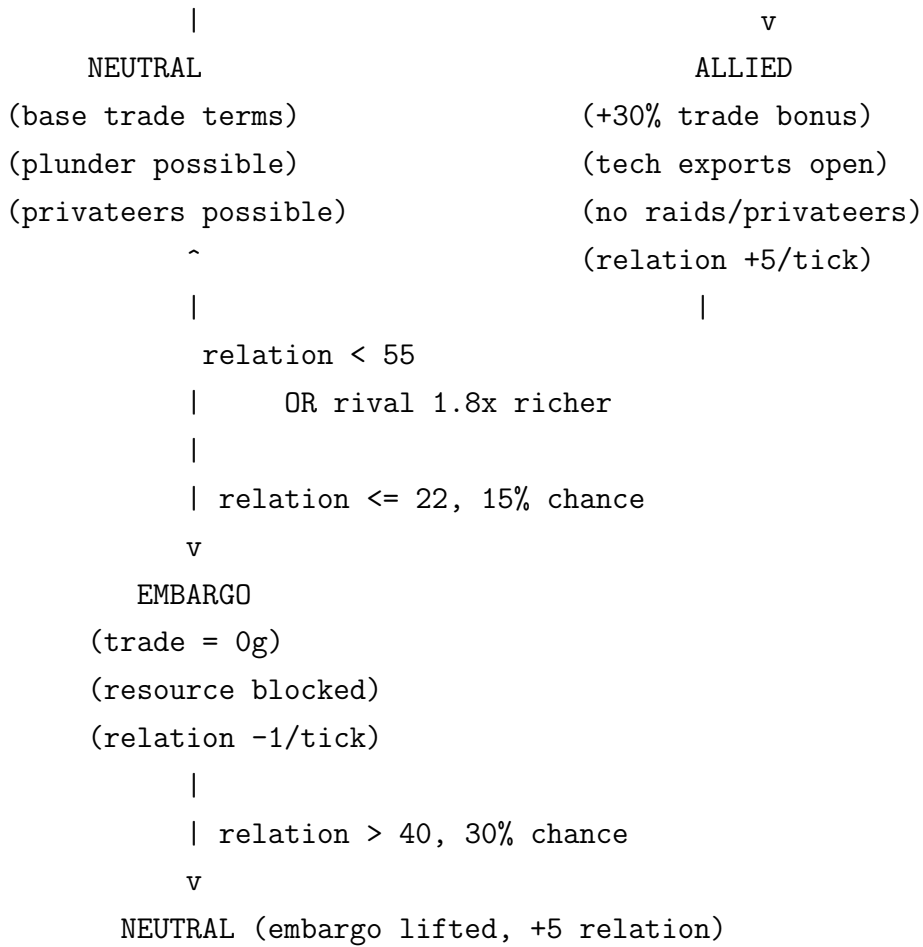
Threshold	Value	Unlocks
Alliance formation	≥ 75 (or ≥ 60 if needing partner)	Commercial treaty
Embargo declaration	≤ 22	Trade embargo
Privateer commission trigger	< 30	Corsair commissioning
Tariff trigger	rival wealth $> 1.15 \times$ own AND rel < 60	Tariff imposition

6.4 Diplomatic States and Transitions

Each pair has one state: **NEUTRAL**, **ALLIED**, or **EMBARGO**. The states are mutually exclusive and symmetric. Diplomatic AI evaluates every 2 ticks starting from tick 3 (mercantilist

scenario only).

relation ≥ 75 , 35% chance



7 MILITARY SYSTEMS

7.1 Naval Fleet

Table 10: Fleet Parameters

Type	Start	Max	Cost/tick
Trade ships	2	unlimited*	0g
Warships	1	4	30g each

*Trade ships can be lost to storms, sabotage, or degradation; minimum is always 1.

7.2 Plunder (Mercantilist Scenario Only)

For each warship of the attacker, for each non-allied victim:

```
if random() <= 0.20 AND victim.tradeShips > 0:
    amount = min(50g, victim.wealth)
    victim.wealth -= amount
    attacker.wealth += amount
```

Allied nations are never raided. Expected plunder per warship per tick:

$$3 \text{ victims} \times 0.20 \times 50\text{g} = 30\text{g gained} \quad (3 \text{ attackers} \times 0.20 \times 50\text{g} = 30\text{g lost, on average}) \quad (6)$$

This confirms plunder is **zero-sum redistribution**, not wealth creation.

7.3 Arms Race (Richardson Model)

The target naval cost for nation i follows the Richardson mutual-reaction model:

$$C_{\text{target}} = \alpha \times C_{\text{max.rival}} + \beta = 0.70 \times C_{\text{max.rival}} + 20 \quad (7)$$

With $\alpha = 0.70 < 1$, the arms race is **convergent** — it does not escalate indefinitely. At symmetric start (all 1 warship, 30g cost), the target $= 0.70 \times 30 + 20 = 41\text{g} \rightarrow 1$ warship, so no escalation occurs from the initial equilibrium.

In the **free trade scenario**, naval cost is fixed at **20g/tick** (beta only, no arms race).

7.4 Privateers vs. Regular Plunder

Table 11: Comparison: Regular Plunder vs. Privateers

Property	Regular Plunder	Privateers
Source	Warships (permanent)	Commissioned corsairs (one-time)
Success rate	20%	55% (77% vs. embargoed)
Amount per hit	50g	65g
Cost	30g/tick maintenance	120g upfront per corsair
Targets	All non-allied	All non-allied

7.5 Sabotage

A nation attempts harbour sabotage when: $\text{wealth} \geq 210g$, the target has the worst relationship score below 40, and the target holds more than 1 trade ship. Attempt probability is **28%** per tick at a cost of 70g win or lose.

Table 12: Sabotage Outcome Table

Outcome	Chance	Effect on Target	Relation Change
Success	55%	Loses 1 trade ship	Aggressor -8 , Victim -30
Failure (uncovered)	45%	No ship loss	Aggressor -8 , Victim -15

8 DEGRADATION & RECOVERY

8.1 Degradation Levels

Checked at step 15 of each tick, based on current wealth:

Table 13: Degradation Level Effects

Level	Wealth Range	Export Penalty	Fleet Effects
Healthy	$\geq 600g$	$\times 1.00$	None
Struggling	250–599g	$\times 0.80$	Lose 1 trade ship after 2 consecutive negative ticks
Critical	100–249g	$\times 0.55$	50% lose warship; 40% lose trade ship
Bankrupt	$< 100g$	$\times 0.30$	65% lose trade ship; 65% lose warship

Trade ships cannot drop below 1 (minimum fleet). Each level re-evaluates **every tick** based on current wealth. Recovery is possible in a single tick via a Gold Rush event or foreign aid transfer.

8.2 Allied Foreign Aid (Recovery Mechanic)

After degradation is applied, if a nation is **critical** or **bankrupt**:

for each allied nation with wealth $\geq 500g$:

aid = floor(donor.wealth * 0.10)

donor.wealth -= aid

recipient.wealth += aid

relations(both) += 8

This breaks the death spiral: wealthy allies act as economic insurance, creating a strategic incentive to form alliances *before* a crisis.

8.3 Degradation State Machine

```

wealth >= 600
|
v
HEALTHY
(export x1.0)
|
| wealth < 600
|
v
STRUGGLING
(export x0.8)
(lose trade ship
after 2 neg. ticks)
|
| wealth < 250
|
v

```

```
CRITICAL
(export x0.55)          < allied aid kicks in
(50% lose warship)
(40% lose tradeship)
    |
    | wealth < 100
    v
BANKRUPT
(export x0.30)
(65% lose both)
```

9 TARIFF SYSTEM

9.1 Imposition (Mercantilist Only, Tick ≥ 2)

A nation imposes a tariff on a rival when **both** conditions hold:

1. $\text{rival.wealth} > \text{nation.wealth} \times 1.15$ (rival has 15%+ more wealth)
2. $\text{getRelationFrom}(\text{self}, \text{rival}) < 60$ (relations are strained)

This models the historical mercantilist response: protect domestic industry when falling behind a wealthier competitor. Before v5.3, tariffs used a trade-balance trigger that could never fire; this was corrected to the wealth-rivalry trigger.

9.2 Tariff Effects and Retaliation

- The imposing nation's import costs **decrease** (beneficial)
- The partner's export income on this route drops by **50%**
- Relations: imposer -3 , target -12 (asymmetric)
- The victim **retaliates after 1 tick**, creating a tariff war cascade

Tick N: Nation A imposes tariff on B (B relation drops -12)

Tick N+1: Nation B retaliates on A

Both nations lose 50% of mutual trade income

Tariffs are **not automatically removed** once imposed; they persist until the simulation ends or removal logic is added.

10 RANDOM MARKET EVENTS

A **40% chance** exists each tick that one random event fires on one randomly selected nation. Over 24 ticks, each nation can expect approximately 2.4 events on average. Events fire at step 5, before trade, and can therefore affect the same tick's calculations.

Table 14: Random Market Events

Event	Effect
Gold Rush	+250g windfall
Plague	$-\min(200g, 30\% \text{ of wealth})$
Great Storm	Lose 1 trade ship; if only 1 ship, lose 15% wealth instead
Market Boom	+25% of current wealth
Market Crash	-25% of current wealth
Mutiny	Lose 1 warship (if > 1)
Trade Wind	Gain 1 trade ship (if < 5); else +100g

11 SCENARIO COMPARISON

11.1 Mercantilist Scenario

All systems are active. Nations impose tariffs, raid each other, form and break alliances, commission privateers, and conduct sabotage. The arms race drives permanent naval costs. Global wealth is lower because:

- Naval maintenance costs (30g/warship/tick) consume productive wealth
- Tariffs and embargoes reduce trade volume for both parties
- Plunder redistributes wealth in zero-sum fashion rather than creating it

Expected final global wealth: $\approx 9,000\text{--}10,000\text{g}$

11.2 Free Trade Scenario

A cooperative environment with no tariffs, plunder, privateers, sabotage, or diplomatic AI. All trade routes remain open with a $\times 1.65$ free trade bonus. Naval cost is fixed at 20g/tick.

Expected final global wealth: $\approx 11,000\text{--}12,000\text{g}$ ($\approx 15\text{--}25\%$ more)

11.3 Academic Result

$$W_{\text{global}}(\text{free trade}) > W_{\text{global}}(\text{mercantilist}) \quad (8)$$

This validates Adam Smith's thesis: free trade is positive-sum (both parties gain), while mercantilism is largely zero-sum (plunder and tariffs redistribute rather than create wealth). The Gini coefficient is also computed: mercantilism typically shows higher inequality (one nation wins big via plunder or tier-3 advantage) while free trade distributes wealth more evenly.

12 DATA COLLECTION

12.1 Per-Tick CSV Schema

scenario, tick, nation, wealth, wealth_delta, exports, imports, plunder_gained, plunder_lost, navy_cost, trade_ships, warships, degradation_level, tariff_count, alliance_count, embargo_count, privateer_count, economy_tier

Note on wealth_delta: This column records $X - M + P_{\text{gained}} - C$. It does **not** include plunder_lost. Actual net wealth change = wealth_delta - plunder_lost.

12.2 Event Log CSV Schema

scenario, tick, event_type, actor, target, detail, amount

Event types: random_event, plunder, alliance_formed, embargo_declared, alliance_broken, embargo_lifted, privateer_commissioned, privateer_raid, sabotage_success, sabotage_failed.

12.3 Computed Aggregates

Table 15: Data Aggregates Computed Post-Simulation

Aggregate	Description
Tariff-wealth correlation	Avg wealth delta with vs. without active tariffs
Alliance trade impact	Avg exports with vs. without active alliances
Arms race burden	Total navy cost as % of total exports
Gini coefficient	Wealth inequality (0 = equal, 1 = one nation holds all)
Scenario comparison	Global wealth, avg wealth, total exports, total navy cost
Per-nation final wealth	Ranked comparison across both scenarios

13 CONSTANTS REFERENCE

13.1 Simulation & Nation Constants

Table 16: Core Simulation Constants

Constant	Value	Description
INITIAL_WEALTH	1,000g	Starting treasury
TICK_DURATION	12s	Real seconds per tick
MAX_TICKS	24	Ticks per scenario

13.2 Resource Constants

Table 17: Resource System Constants (v5.3)

Constant	Value	Description
RESOURCE_MIN_NEED	15	Urgent buy threshold (was 20)
RESOURCE_MAX_STOCK	200	Saturation cap (was 100)
RESOURCE_START_STOCK	40	Starting stock per resource
RESOURCE_OWN_PRODUCTION	20	Own production per tick (was 15)
RESOURCE_CONSUMPTION	4	Consumption per resource per tick (was 8)
RESOURCE_URGENT_PREMIUM	1.5	Premium when stock < MIN_NEED
RESOURCE_TRADE_AMOUNT	6	Units shipped per ship per route (was 12)

13.3 Trade Constants

Table 18: Trade System Constants (v5.3)

Constant	Value	Description
BASE_EXPORT_INCOME	160g	Total base income split across routes (was 95)
BASE_ROUTE_INCOME_FLOOR	10g	Minimum per active route (new in v5.2)
IMPORT_SPEND_RATIO	0.55	Imports = 55% of exports
FREE_TRADE_BONUS	0.65	Bonus multiplier in free trade scenario (was 0.40)
ALLIANCE_TRADE_BONUS	0.30	+30% on allied routes

13.4 Military Constants

13.5 Relationship Constants

Table 19: Military Constants (v5.3)

Constant	Value	Description
INITIAL_WARSHIPS	1	Starting warships
WARSHIP_COST_PER_TICK	30g	Maintenance per warship
MAX_WARSHIPS	4	Cap (was 8)
PLUNDER_SUCCESS_RATE	0.20	Raid chance per warship (was 0.45)
PLUNDER_AMOUNT	50g	Gold per successful raid (was 85)
ARMS_RACE_ALPHA	0.70	Richardson coefficient (was 1.25; now < 1)
ARMS_RACE_BETA	20	Richardson constant (was 15)

Table 20: Relationship Constants (v5.3)

Constant	Value	Description
RELATION_INITIAL	50	Neutral starting score
RELATION_ALLIANCE_THRESHOLD	75	Score to trigger alliance
RELATION_EMBARGO_THRESHOLD	22	Score to trigger embargo (was 25, then 10)
RELATION_RAID_THRESHOLD	30	Privateer trigger (was 35)
RELATION_DECAY_RATE	1	Drift/tick toward 50 (was 3 in v5.2)
RELATION_TRADE_BOOST	+3	Per successful trade (was 2)
RELATION_ALLIANCE_BOOST	+5	Per allied tick
RELATION_EMBARGO_DRAIN	-1	Per embargoed tick (was -3)
RELATION_RAID_VICTIM	-10	Victim penalty per raid (was -25)
RELATION_RAID_AGGRESSOR	-3	Attacker penalty (was -5)
RELATION_TARIFF_VICTIM	-12	Tariff target resentment

14 ACADEMIC FRAMEWORK

14.1 Economic Theories Modeled

Table 21: Academic Sources per Game Mechanic

Mechanic	Theory	Source
Free trade raises global wealth	Positive-sum trade / comparative advantage	Smith (1776); F (1817)
Tariff → retaliation cascade	Trade war / Nash equilibrium	Nash (1950); Smoot-L (1930)
Arms race formula	Richardson mutual-reaction model	Richardson (1960)
Alliance / embargo / privateer AI	Iterated Prisoner’s Dilemma	Axelrod (1984)
Resource dependency diplomacy	Interdependence theory	Keohane & Nye (197)
Gini inequality measurement	Wealth distribution	Gini (1912)

14.2 Version History

Table 22: Key Changes Across Versions

Version	Key Fix	Impact
v4.0	Bilateral relationship system	Diplomacy driven by scores, not wealth
v4.1	Resource-need diplomacy	Alliance/embargo shaped by supply dependency
v5.0	Allied foreign aid; ally-safe raids; asymmetric relations	Death spiral recovery; realistic diplomacy
v5.1	ALPHA < 1; plunder rate/amount halved; MAX_WARSHIPS 8 → 4	Stopped universal bankruptcy by tick 8
v5.2	Resource balance (CONSUMPTION 8 → 4); route floor income	Stopped free trade income collapse by tick 5
v5.3	Tariff trigger fixed (wealth rivalry); EMBARGO_THRESHOLD 10 → 22; free trade bonus 0.40 → 0.65; nation names updated	Diplomacy fires correctly; academic result confirmed

15 CONCLUSION

This report has documented the complete economic model of the Mercantilism Simulation (v5.3), covering all subsystems that govern how the four IT-sector nations — **Endava**, **Amdaris**, **GridDynamics**, and **Globant** — accumulate, transfer, and lose wealth over 24 trading ticks.

The wealth formula $W_i(t) = W_i(t - 1) + X_i - M_i + P_i - C_i$ drives all outcomes, with each term shaped by the relationship, diplomacy, tariff, naval, and degradation systems operating in concert.

The mercantilist scenario activates all competitive mechanisms: arms races, plunder, tariff wars, and embargoes. While individual nations can gain through plunder or tier-3 technology exports, global wealth is reduced by naval maintenance costs and trade disruption. Plunder, being zero-sum, merely redistributes the existing stock of gold.

The free trade scenario removes all competitive disruptions and applies a $\times 1.65$ income bonus to all routes. The result is consistently higher global wealth ($\approx 15\text{--}25\%$ more), validating Adam Smith’s positive-sum thesis and Ricardo’s comparative advantage framework.

The version history reflects an iterative design process in which each parameter change was motivated by emergent failures: universal bankruptcy from an overpowered arms race, income collapse from excessive consumption rates, and diplomatic silence from a broken tariff trigger. Version 5.3 represents a stable, academically valid configuration where both scenarios produce meaningful and contrasting outcomes.

The data collection infrastructure (per-tick CSV and event log) enables post-hoc analysis of all aggregates, from the arms race burden to the Gini coefficient, allowing empirical verification of the simulation’s theoretical predictions.